

Teorema 1. *Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ injective, $a \in {}^*\mathbb{R} \setminus \mathbb{R}$. Then $f(a) \in \mathbb{R} \setminus \mathbb{R}$.*

Proof. By the choice of the fol, we already have f is a function symbol of our language. This guarantees that $f(a)$ makes sense. Let $r = f(a)$, that is ${}^*\mathbb{R} \models f(a) = r$. By contradiction, let's say $r \in \mathbb{R}$. Since $\mathbb{R} \models f$ injective, the same goes for ${}^*\mathbb{R}$. Since ${}^*\mathbb{R} \models \exists x f(x) = r$, by elementarity the same goes for $\mathbb{R}$. Let $d \in \mathbb{R}$ be such that $\mathbb{R} \models f(d) = r$. Then ${}^*\mathbb{R} \models f(d) = r \wedge f(a) = r$. This is a contradiction since ${}^*\mathbb{R} \models f$ injective and $a \neq d$ since $a \notin \mathbb{R}$ while $d \in \mathbb{R}$.

□